Evaluating Multimodal Fusion Robustness in Drone-Based Object Detection Under Sensor Degradation

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Abstract

Autonomous drones deployed in safety-critical applications rely on multimodal detection systems that fuse heterogeneous sensor streams. These systems are routinely evaluated on clean data, leaving their behavior under real-world sensor degradation poorly understood and researched. This research systematically applies a corruption suite of nine degradation types (including Gaussian noise, motion blur, brightness shift etc) at three calibrated severity levels. These degradations are applied to the DroneVehicle RGB-IR dataset, evaluating multimodal fusion model performance using the Resistance Ability metrics adapted from the MultiCorrupt benchmark. Through targeted modality removal experiments, we also quantify the contribution of each sensor stream under specific corruption conditions which provides concrete guidance on which modality to prioritize when designing fault-tolerant drone detection systems.

Keywords

drone perception, sensor fusion, multimodal object detection, robustness evaluation, sensor corruption, thermal infrared imaging, RGB-IR fusion

1 Introduction

Multimodal object detection on UAV platforms typically fuses heterogeneous sensor streams to compensate for the limitations of each sensor individually. Visible light cameras degrade in low-light or smoke conditions, while thermal sensors lose discriminability when target and background temperatures converge. Even then, the published models are almost exclusively benchmarked on uncorrupted data, which does not reflect real-world operational conditions. In practice, sensors are subject to motion blur from platform vibration, radiometric noise from electronic interference, brightness artifacts from direct solar exposure, and complete modality dropout from hardware failures. The absence of systematic robustness evaluation means that the mAP figures reported in the literature cannot predict how a model performs when one or both sensors are compromised.

This proposal addresses that gap directly. We apply a nine-type corruption suite at three severity levels to the DroneVehicle RGB-IR dataset and evaluate three fusion architectures using Resistance Ability (RA) metrics adapted from the MultiCorrupt benchmark. Two research questions structure the work: RQ1 asks how detection performance degrades under specific corruption types and severities, RQ2 asks which modality contributes most to robustness under each degradation condition. The remainder of the proposal details the dataset, corruption pipeline, model selection, evaluation protocol, and project timeline.

2 Problem Statement

Current RGB-IR fusion models for drone-based detection have no standardized way to measure robustness degradation. Metrics such as mAP quantify detection accuracy on a fixed test set but say nothing about how performance scales with corruption severity or whether degradation is uniform across fusion strategies. MultiCorrupt [3] established this kind of structured evaluation for LiDAR-camera fusion on nuScenes, but no equivalent exists for RGB-thermal drone detection. The architectural question is similarly open: Ma et al. [1] showed that fusion strategy determines robustness under missing modality, but the result was dataset dependent and tested on vision language classification, not sensor fusion for object detection.

2.1 Research Questions

RQ1: How does the detection performance of a multimodal fusion model degrade under different types and severity levels of sensor corruption (noise, blur and missing modality) in drone-based object detection?

RQ2: Which sensor modality, RGB or thermal infrared, contributes the most to model resilience under specific degradation types, and which is most critical to preserve?

RQ1 is diagnostic as it maps the degradation profile across nine corruption types and three severity levels for each fusion architecture. RQ2 is more attributional where modality removal under each corruption condition isolates which sensor carries detection performance when the other is degraded. Together they produce a robustness characterization that mAP alone cannot provide.

3 Related Work

Multimodal fusion robustness. Ma et al. [1] tested Transformer-based multimodal models on MM-IMDb, UPMC Food-101 and Hateful Memes with progressively removed modalities. Performance dropped sharply under partial modality absence. On MM-IMDb, multimodal accuracy fell below the unimodal baseline when text availability dropped to 30%. More relevant to this work: the optimal fusion strategy is data dependent. Early fusion outperformed late fusion on MM-IMDb and Hateful Memes, but late fusion was better on Food-101. The implication is that fusion strategy selection cannot be made on theoretical grounds alone, it requires empirical evaluation on the target domain.

Robust multimodal detection architectures. Chu et al. [2] proposed MT-DETR for object detection in adverse weather using camera, lidar, radar and time inputs. The Confidence Fusion Module (CFM) computes a confidence map from concatenated multi-modal features, multiplies it element wise against the depth feature, and adds the results to the camera branch. This lets the model learn which sensor to trust per input rather than using fixed fusion weights set during training. MT-DETR was evaluated on the STF dataset across clear, fog and snow conditions, outperforming unimodal and prior fusion baselines. The CFM design introduced the confidence-weighted fusion concept but is superseded in this study by C^2Former [7], which applies cross-attention at the feature level rather than confidence-map gating.

Corruption benchmarking methodology. Beemelmanns et al. [3] introduced MultiCorrupt, a systematic robustness benchmark for LiDAR-camera fusion on nuScenes. Ten corruption types at three severity levels are applied independently to each modality, and model performance is evaluated using Resistance Ability, normalized across all models. The benchmark showed that no single fusion architecture was robust across all corruption types, LiDAR-dominant models performed better under camera corruptions, while camera-dominant models degraded less under LiDAR noise. No equivalent benchmark exists for RGB-thermal fusion on drone platforms, where the sensor characteristics, failure modes and scene geometry differ substantially from automotive data.

Infrared corruption benchmarking. Medeiros et al. [6] introduced LLVIP-C and FLIR-C, two corruption benchmarks applied directly to infrared object detection datasets using standard image-space transforms at five severity levels. Their work confirms that TIR models are sensitive to noise and blur corruptions and that robustness varies across detector architectures. The key distinction from this study is scope: LLVIP-C and FLIR-C evaluate single-modality IR detectors, not RGB-IR fusion models, and neither dataset is drone-based. No equivalent benchmark evaluates how RGB-thermal fusion degrades under joint per-modality corruption on aerial data.

4 Methodology

4.1 Dataset

Experiments use the DroneVehicle dataset [4], which contains 28,439 synchronized RGB and thermal infrared image pairs captured from UAV platforms over urban and suburban scenes across day and night. Annotations cover five vehicle classes; car, truck, bus, van and freight car. These annotations use oriented bounding boxes. The two modalities are captured independently by separate sensors mounted on the same platform, which means corruptions can be applied to each input stream without affecting the other. This test split is used exclusively for evaluation, no fine-tuning is performed. Unlike automotive multimodal datasets such as nuScenes, DroneVehicle does not include LiDAR data, as the UAV platforms used for collection do not carry LiDAR sensors. The study is therefore scoped to RGB-TIR fusion, which is the sensor configuration available on this class of drone.

4.2 Corruption Design

The corruption pipeline is adapted from the MultiCorrupt codebase [3] and uses the `imagecorruptions` Python library [5] for all image-space transforms. The library supports grayscale inputs natively, so the same corruption functions apply to both the RGB and TIR channels without modification. RGB corruptions (Gaussian noise, motion blur, brightness shift, and low contrast) and TIR corruptions (sensor noise, blur, and intensity shift) are each applied at severity levels 1, 2, and 3. Intensity shift for TIR is implemented via the library's `brightness` function applied to the grayscale channel, using the same additive HSV offset parameterisation as the RGB brightness shift corruption. This approach follows the precedent set by Medeiros et al. [6], who applied the same library to TIR images to construct the LLVIP-C and FLIR-C infrared corruption benchmarks. Complete dropout for each modality is treated as a binary condition with no severity levels. All corruptions are applied to the test split only. Severity levels are parameterised using the library's built-in scale: for Gaussian noise, severity controls the standard deviation σ of the additive noise; for motion blur, it controls kernel radius and σ; for brightness shift, it is an additive offset in HSV value space; and for contrast, it is a multiplicative scaling factor. Full parameters per corruption type and severity level are listed in Table 2. The full corruption matrix is shown in Table 1.

Table 1: Corruption suite applied per modality and severity level.

Corruption Type	Modality	Severity
Gaussian Noise	RGB	1 / 2 / 3
Motion Blur	RGB	1 / 2 / 3
Brightness Shift	RGB	1 / 2 / 3
Low Contrast	RGB	1 / 2 / 3
Complete Dropout	RGB	—
Sensor Noise	Thermal (IR)	1 / 2 / 3
Blur	Thermal (IR)	1 / 2 / 3
Intensity Shift	Thermal (IR)	1 / 2 / 3
Complete Dropout	Thermal (IR)	—

Table 2: Severity level parameters for each corruption type.

Corruption	Sev. 1	Sev. 2	Sev. 3
Gaussian Noise	$\sigma = 0.08$	$\sigma = 0.12$	$\sigma = 0.18$
Motion Blur	$r = 10,\ \sigma = 3$	$r = 15,\ \sigma = 5$	$r = 15,\ \sigma = 8$
Brightness Shift	+0.10	+0.20	+0.30
Low Contrast	0.4×	0.3×	0.2×
Sensor Noise (TIR)	$\sigma = 0.15$	$\sigma = 0.20$	$\sigma = 0.35$
Blur (TIR)	$\sigma = 1$	$\sigma = 2$	$\sigma = 3$
Intensity Shift	+0.10	+0.20	+0.30

σ: noise std dev or blur kernel std dev; r: blur kernel radius.
All values normalised to $[0, 1]$ pixel range.

4.3 Modality Removal

For each corruption condition, two additional inference runs are performed alongside the full dual-modality evaluation: one with the RGB stream zeroed out and one with the TIR stream zeroed out. Comparing full-modality mAP against each single-modality result under the same corruption isolates which sensor carries detection performance when the other is degraded. This directly addresses RQ2.

4.4 Models and Baselines

Three fusion architectures are trained on DroneVehicle using the EEMCS HPC cluster and evaluated under corruption. All three use ResNet-50 initialised from ImageNet weights.

UA-CMDet [4] is the primary baseline. The uncertainty-aware RGB-IR fusion model from the DroneVehicle paper. It processes each modality through separate branches and combines them using uncertainty-guided feature weighting. Clean-data mAP is verified against the published figures before any corruption is applied, confirming that the pipeline is correct.

The second model is an early-fusion variant where RGB and TIR feature maps are concatenated before the main encoder, following the strategy Ma et al. [1] found most robust under missing modality on two of three tested datasets. It is the architecturally simplest of the three and sets a baseline for how much robustness straightforward concatenation provides.

The third model is C^2Former [7], a cross-attention intermediate fusion model that uses an Inter-modality Cross-Attention (ICA) module to learn calibrated feature correspondences between RGB and IR streams at the backbone level. Unlike concatenation-based approaches, C^2Former dynamically attends to cross-modal feature relationships rather than fusing fixed representations. It is benchmarked on DroneVehicle in the original paper, making published mAP figures available for validation. Across the three architectures, the experiments test whether Ma et al.'s finding transfers to RGB-IR drone detection: that fusion strategy selection requires empirical validation on the target domain and cannot be determined theoretically.

4.5 Evaluation Metrics

Detection performance is measured with mAP at IoU 0.5, computed using DOTA evaluation toolkit to correctly handle oriented bounding box annotations. Robustness is reported using Resistance Ability (RA) which is defined as corrupted mAP divided by clean mAP, this is also adapted from MultiCorrupt [3]. RA is reported per corruption type, per severity level and per modality condition for each of the three fusion architectures.

5 Expected Results

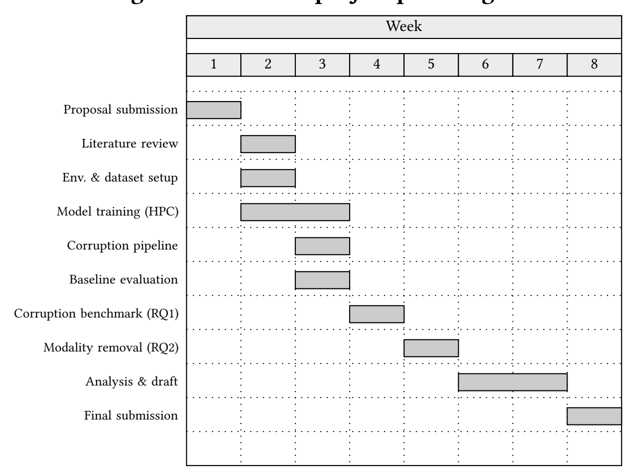
RQ1. For each fusion architecture, the experiments will produce a degradation profile: mAP and RA reported per corruption type and severity level. Complete modality dropout is expected to cause the sharpest mAP drops, since it removed all information from one sensor stream. Among graded corruptions, noise-type degradations on TIR are likely to be more damaging than equivalent RGB corruptions, as WiSE-OD [6] found TIR models are particularly sensitive to noise. RA will likely decrease with severity for most corruption types, but the rate of decline is expected to differ across fusion strategies, which is consistent with MultiCorrupt's finding that no single architecture holds up uniformly across all corruption types [3].

RQ2. The modality removal is expected to show TIR is the more load-bearing sensor under RGB-specific corruptions such as brightness shift and low contrast. This is mainly because thermal imaging is unaffected by visible light degradation. Conversely, RGB is expected to contribute more to the fine-grained vehicle class discrimination, where thermal images lose resolution between target and background at similar temperatures. The practical output is per-corruption modality priority map which gives a concrete guidance on which sensor to weight or preserve when designing fault-tolerant fusion systems for drone deployment. Whether C^2Former's cross-attention mechanism adapts its inter-modal feature alignment dynamically under corrupted inputs is an open question this study is designed to answer.

6 Planning

Figure 1 outlines the project schedule across eight weeks. The timeline assumes proposal approval at the end of Week 1, with experimental work beginning immediately after. Weeks 2 and 3 run the environment setup and the corruption pipeline in parallel with the literature review, since both can proceed independently. The clean baseline evaluation in Week 3 acts as a validation checkpoint before any corruption experiments begin. Corruption benchmarking and modality removal experiments follow sequentially in Weeks 4 and 5 leaving Week 6 through 8 for analysis, writing and final submission.

Figure 1: Research project planning.



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